

DashVMC

Real-Time Discrete World Model Control in Geometry Dash

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World-model agents are usually evaluated in simulators that can wait for the policy; live games impose the opposite constraint, requiring capture, prediction, and action before the next frame. We present *DashVMC*, a discrete world-model agent that plays Geometry Dash from screen pixels. A Finite Scalar Quantization (FSQ) tokenizer compresses Sobel-filtered observations into discrete token grids, a block-causal transformer predicts action-conditioned transitions, and a lightweight actor-critic is initialized by behavioural cloning (BC) and subsequently optimized with Proximal Policy Optimization (PPO) in latent rollouts. Training uses approximately 2 hours of gameplay captured at a nominal 30-FPS cadence, including less than 20 minutes of expert segments; all learning is offline, without any real-environment interaction by the agent. Across three independent controller seeds, PPO has higher mean live survival than its exact retained parent BC on every official level for every seed, with the largest gains on the first two levels. Deployed locally, the complete capture-to-action pipeline supports reactive control at 60 FPS on a consumer GPU. Beyond control, the learned dynamics generate interactive, open-ended visual level continuations that can remain plausible over extended rollouts. These results demonstrate that a specialized world model can jointly support imagined policy optimization, interactive generation, and real-time deployment.

Project page: tardieu.github.io/dash-vmc

Code: github.com/Tardieu/dash-vmc

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1 Introduction

A world model learns an environment’s transition dynamics, conventionally written $p_\theta(s_{t+1} | s_t, a_t)$: given a state s_t and action a_t , it predicts the state that follows (LeCun, 2022). When privileged environment state is unavailable, as here, the model instead operates on a learned latent representation inferred from recent observations and actions. Rolling the model forward allows an agent to improve its policy in imagination rather than through additional interaction with the real environment (Ha and Schmidhuber, 2018). Discrete approaches such as IRIS encode images into tokens and model their evolution with an autoregressive transformer (Michele et al., 2023), whereas DIAMOND uses diffusion to preserve fine visual detail (Alonso et al., 2024). These designs do not inherently preclude real-time action inference because their policies need not generate a new world-model observation while acting in the real environment. DashVMC instead targets a joint systems requirement: inexpensive imagined transitions for policy optimization together with a decoder-free live path that reuses the dynamics model’s temporal state. It follows IRIS’s discrete autoregressive formulation across frames, but predicts each next token grid in parallel and is adapted to an approximately two-hour offline corpus and a live-control setting.

Regardless of model class, most world-model agents are evaluated in synchronous, step-based environments whose simulation advances only when the agent submits an action. Controlling a live, non-paused application imposes an additional wall-clock constraint: screen capture, preprocessing, state estimation, policy inference, and action dispatch must finish before the action becomes stale.

Geometry Dash is a small but unforgiving test of this contract. The avatar moves continuously, the action space is binary, and a mistimed jump causes immediate death. Levels are deterministic, but the agent observes only captured pixels and acts through keyboard events; it does not receive emulator state during control. This combination makes Geometry Dash a practical deployment stress test: can a pixel-based world-

model agent close the live capture-to-action loop quickly and precisely enough for narrow timing windows on a consumer GPU?

DashVMC adopts the broad perception–dynamics–control organization introduced by World Models (Ha and Schmidhuber, 2018). An FSQ tokenizer (Mentzer et al., 2024) encodes Sobel edge maps. An action-conditioned transformer world model predicts the next token grid from the recent latent context and a chosen action, enabling autoregressive imagined rollouts of the controller’s decisions. The controller receives both the current observation grid and the transformer’s temporal state; after a short BC warm start, it is optimized by PPO (Schulman et al., 2017) entirely in rollouts generated by the frozen world model. At deployment, neither decoding nor next-frame prediction is executed: the encoder, transformer context pass, and controller run once per frame, while policy optimization has already happened offline.

This is a systems co-design contribution rather than a new generic world-model primitive. The 64-token visual grid bounds live perception and context-encoding cost, while the parallel next-frame predictor bounds the cost of imagined rollouts; explicit action conditioning permits counterfactual rollouts; BC avoids an expensive random-policy transient; and GPU-resident state plus captured execution remove avoidable synchronization from the live path. Our contributions are:

1. an end-to-end discrete world-model agent for pixel-based control of the original, non-paused Geometry Dash game, whose parallel imagined transition fits within a 30-FPS frame period and whose decoder-free capture-to-action path sustains a 60-FPS live cadence on a consumer GPU; and
2. an offline training pipeline using approximately two hours of captured gameplay, including less than 20 minutes of expert segments, that initializes a lightweight controller with BC and optimizes it with PPO entirely in autoregressive rollouts of the frozen action-conditioned world model, without any real-environment interaction by the agent during learning.

We evaluate these contributions through three retained BC-to-PPO controller pairs on three official and two auxiliary layouts, matched-action interventions, end-to-end latency profiling, and qualitative open-ended rollouts and failure analysis.

2 Related work

World models and imagination. The original Vision–Memory–Controller pipeline compressed observations with a VAE, modeled latent dynamics with an MDN–RNN, and demonstrated in VizDoom that a compact controller could be optimized entirely in learned dreams and transferred to the real environment (Ha and Schmidhuber, 2018). Subsequent imagination-based agents explore different representations and dynamics models: IRIS, the closest architectural reference for DashVMC, combines a discrete autoencoder with an autoregressive transformer for data-efficient Atari learning (Micheli et al., 2023); Δ -IRIS separates stochastic changes into discrete tokens from continuous context tokens to reduce sequence cost (Micheli et al., 2024); and DIAMOND uses diffusion without discrete compression for imagination-trained control and also demonstrates an interactive Counter-Strike neural game engine (Alonso et al., 2024). DreamerV3 demonstrates broad control with categorical recurrent state-space latents (Hafner et al., 2025), while TWISTER augments transformer world-model training with action-conditioned contrastive predictive coding to learn temporal representations over longer horizons (Burchi and Timofte, 2025). At larger generative scales, related action-conditioned simulators include MIRA, which models tightly coupled four-player Rocket League dynamics with latent diffusion (Hu et al., 2026); MineWorld, which autoregressively models interleaved discrete visual and action tokens with parallel spatial decoding (Guo et al., 2025); and Matrix-Game 2.0, which uses few-step diffusion for streaming generation (He et al., 2025). Within Geometry Dash specifically, earlier screenshot-based control used direct DQN and imitation learning under a DLL-fabricated timestep, and highlighted that narrow jump windows can make nearly identical frames require different actions (Li and Rafferty, 2017). Our protocol leaves the game clock running and supplies the controller with temporal context through the learned world model.

FSQ tokenization and policy initialization. FSQ replaces learned vector-quantization codebooks with bounded scalar levels, avoiding commitment losses and codebook collapse while exposing an integer coor-

dinate lattice (Mentzer et al., 2024); we use the improved iFSQ bounding function (Lin et al., 2026). For policy initialization, demonstration-based RL has repeatedly shown that supervised behaviour can reduce the expensive early interaction phase (Hester et al., 2018; Baker et al., 2022). Here demonstrations only initialize the controller: the final policy is selected after PPO improvement in the learned environment.

Representation anchors and alternatives. Joint-embedding prediction can avoid spending capacity on high-magnitude input detail that is irrelevant or unpredictable from context (Van Assel et al., 2025), and LeWorldModel demonstrates stable end-to-end predictive learning from pixels with continuous latents (Maes et al., 2026). DashVMC instead applies deterministic preprocessing that converts each cropped RGB capture to grayscale and then to a Sobel edge map, removing much appearance detail before learning and making reconstruction a comparatively inexpensive anchor for the shared discrete token space. Among discrete tokenizers, Gaussian Quant reports stronger rate-distortion performance than FSQ on large image benchmarks (Xu et al., 2025). Whether those gains transfer to low-resolution Sobel edge maps and downstream world-model control remains untested.

3 DashVMC

The three modules are trained sequentially: V and M are frozen before controller fitting, and the selected controller is frozen for deployment. Figure 1 separates the imagined PPO path from the live reactive path.

3.1 Visual tokenization

Each RGB capture is cropped, converted to grayscale, filtered with Sobel edges, and resized to 64×64 . A convolutional autoencoder maps the result through three stride-2 stages to a four-channel 8×8 latent map. Each spatial vector is independently quantized with iFSQ levels $[8, 5, 5, 5]$, using the bounding function $2 \text{sigmoid}(1.6z) - 1$, and yields 64 token IDs from 1000 implicit codes. The tokenizer is trained on adjacent frame pairs with pixel-MSE reconstruction plus temporal slowness and latent-uniformity regularizers (Xia et al., 2025). The encoder and decoder contain 941,924 and 941,921 trainable parameters, respectively. After selection, the encoder is frozen and the decoder is retained only for visualizing dreams.

3.2 Action-conditioned dynamics

The dynamics model has 8 transformer layers, width 384, and 8 attention heads, for 14,723,520 unique trainable parameters including 141,120 in the contrastive predictive coding (CPC) training heads. Each frame block contains 64 visual tokens and one ALIVE/DEATH status token; action tokens are inserted between blocks. Given $K = 4$ observed frame-action pairs, a final masked block requests the next frame. The resulting sequence has $K(65 + 1) + 65 = 329$ positions; every target-frame position receives a learned [MASK] embedding, so no target token is exposed during training. Attention is bidirectional within a frame and causal across frame/action blocks, so all next-frame positions are predicted in one forward pass without seeing their targets. Factored 3D rotary embeddings encode row, column, and time, with spatial and temporal bases 10 and 10,000; status and action tokens use the grid centre coordinates (Su et al., 2024). The status logits supply the rollout termination score, and the final context state $\mathbf{h}_t \in \mathbb{R}^{384}$ summarizes recent dynamics for the controller. In the reported interleaved sequence, this state is taken from the most recent action-token position. Death frames are oversampled $5\times$ during training to prevent the overwhelmingly common ALIVE class from admitting a trivial always-alive solution. Consequently, the DEATH score from a two-class softmax over ALIVE/DEATH is used operationally rather than interpreted as a calibrated environment probability.

We add TWISTER’s action-conditional contrastive predictive-coding loss with weight 0.1 (Burchi and Timofte, 2025): for each future offset $k \leq K$, an MLP predicts $\mathbf{h}_{t+k}$ from $\mathbf{h}_t$ and the intervening actions and is scored with in-batch InfoNCE (van den Oord et al., 2018). We also independently replace 5% of context tokens uniformly and 5% with adjacent FSQ codes. The token loss uses focal modulation (Lin et al., 2017) and structured label smoothing (SLS), implemented as a fixed local FSQ-coordinate target. For target code

3.3 Controller and latent policy improvement

The 45,546-parameter actor-critic embeds the current 8×8 token grid, processes it with two convolutions and max pooling, concatenates the resulting 256-dimensional feature with $\mathbf{h}_t$, and uses zero-initialized heads to predict jump probability, value, and an eight-action auxiliary output covering the current and next seven actions. The controller is first fit to expert actions by BC, then optimized with PPO in autoregressive rollouts from the frozen world model.

Each PPO iteration starts from recorded four-frame contexts and generates 512 imagined episodes of at most 45 steps. The controller acts from the first imagined step, but the first four transitions form a grace period that is excluded from PPO updates. At this short horizon, deaths are usually already determined by the recorded approach state—for example, a jump may be ending too close to an obstacle for any action to prevent the collision—so the exclusion keeps such effectively unavoidable outcomes from influencing the policy update. These four transitions also refresh the rolling context, so retained transitions use an entirely model-generated four-frame context. The 45-step horizon roughly covers obstacles visible in the recorded seed. Beyond it, they have usually left the crop, so the model can enter a physically valid but uninformative empty segment; later obstacles must also be generated rather than seed-observed, progressively shifting controller inputs away from encoded real frames. We stop at this boundary to learn consequences of observed obstacles without relying on invented challenges. Rollouts use the fixed argmax boundary of the two status logits, equivalently terminating when the operational death score exceeds 0.5; this threshold is not interpreted as a calibrated 50% environment probability. Reward is +1 per survived step and -0.25 per jump, preventing an always-jump local optimum. We use Generalized Advantage Estimation (GAE) (Schulman et al., 2015) with $\gamma = 0.995$, $\lambda = 0.95$, PPO clip 0.2, and auxiliary-action loss weight 0.1.

3.4 Reactive deployment path

The training corpus is captured at a nominal 30-FPS cadence, but the optimized live stack is deployed at 60 FPS. Desktop Duplication captures only the 1032×1032 model crop; after CPU grayscale conversion, CUDA Sobel filtering and an ordered area-resize kernel reproduce the recorded-data preprocessing while keeping the normalized 64×64 observation on the GPU. Across 128 synthetic crops and 100 fresh live captures, this path produced zero output-byte mismatches against the original CPU pipeline. Token context remains in a GPU-resident ring buffer. At each step, the transformer runs context prefill only to obtain $\mathbf{h}_t$; its masked next-grid prediction phase is not executed. The packaged live path retains the 941,924-parameter encoder, 14,582,016-parameter transformer context path, and 45,546-parameter controller, for 15,569,486 parameters in total. It discards the 941,921-parameter decoder, 141,120 CPC-head parameters, and 384-parameter mask embedding. The 384,768-entry vocabulary projection is tied to the input token embedding, so removing its prediction-module alias saves no additional weights; skipping masked prediction primarily saves computation and activations. The FSQ encoder uses `torch.compile`, and transformer context encoding is captured in a CUDA Graph after warm-up. A keyboard update is issued only when the binary action changes. The loop sleeps after action dispatch to maintain the selected cadence, but that deliberate sleep is excluded from capture-to-action latency. This is deliberately reactive rather than an inference-time planning system: online search or trajectory optimization would have to share the same 16.7-ms (60-FPS) budget with capture and dispatch.

4 Experimental setup

Data and training. The unaugmented corpus contains 4,264 base gameplay episodes and 251,963 source frames, equivalent to approximately 2.33 hours at the nominal 30-FPS capture cadence. Each recorded frame is paired with the binary keyboard action applied at that timestep (jump or idle), yielding action-labelled trajectories for dynamics learning and BC. It includes deliberate failures and 36 expert trajectories or segments, with the expert subset totaling less than 20 minutes of gameplay. The base episodes span primarily official Levels 1–7 and multiple community-created levels. Episodes containing mechanics outside the selected model’s scope, including gravity inversion, were excluded to keep the learned dynamics tractable at this scale. Episodes are split once, 90/10, by base identifier and source with seed 42, and all derived samples inherit the base episode’s assignment. During FSQ training, each adjacent-frame pair receives a

shared two-dimensional translation whose horizontal and vertical integer offsets are sampled independently and uniformly from $[-4, 4]$ pixels, with border padding. For dynamics training, each episode is tokenized at vertical offsets $\{-4, -2, 0, 2, 4\}$, expanding the corpus to 21,320 episodes. The frozen encoder retokenizes the corpus before dynamics training. Tokenizer and dynamics training use seed 42; the controller replication uses seeds 43, 44, and 45 while holding those two components fixed. Base training is sequential in BF16 on one A100, the controller replications run in BF16 on one H200 at a time, and live inference is measured on an RTX 2060. Table 1 records the optimization and selection details that materially determine the frozen system.

Table 1 Training and checkpoint selection. Tokenizer/dynamics learning rates use cosine decay after any stated warm-up.

Stage	Optimization	Selection
Tokenizer	Adam; 1,000 epochs; batch 2,048; cosine LR from 10^{-3} to 10^{-5} ; slowness 0.1; uniformity 0.01; validation every 10 epochs.	Min. loss (epoch 920).
Dynamics	AdamW; 200 epochs $\times$ 500 steps; batch 512; LR $2 \times 10^{-3} \rightarrow 5 \times 10^{-5}$ after 5-epoch warm-up; weight decay 0.01; dropout 0.1; max grad. norm 1.0; death frames oversampled $5 \times$.	Max. death F1 (epoch 139).
BC	Three seeds; AdamW; 50 epochs/seed; batch 512; LR 10^{-3} ; weight decay 10^{-4} ; positive jump weight 1.5.	Min. loss (10; 10; 11).
PPO	Three seeds; Adam; 15,000 iterations/seed; 512 rollouts/iteration; four update epochs; minibatch 512; LR 10^{-4} ; entropy/critic weights 0.01/0.5; ratio clip 0.2; max grad. norm 0.5.	Best development survival (12,420; 5,090; 12,280).

The historical end-to-end stage logs sum to approximately 12.6 A100 hours. The three reported controller replications add 51.54 H200 wall-clock hours and execute 45,000 PPO iterations in total; per-seed controller times are 17.41, 17.06, and 17.07 hours. These logged times exclude offline preprocessing, live evaluation, and manual handoffs between stages.

Checkpoint and live evaluation. PPO checkpoint selection uses survival on 512 fixed latent development contexts every ten iterations. Because training seeds may originate from the same episode collection and selection is adaptive, this is explicitly a development metric, not held-out validation. For live evaluation, all checkpoints are frozen and Auto-Retry is enabled. Auto-Retry only standardizes restarts and supplies no observation to the policy; the policy receives neither a level identifier, elapsed time or frame index, level progress, nor game memory. Live actions are deterministic: jump is selected exactly when the controller output exceeds 0.5. For each controller seed, we retain 25 scored attempts for both the selected BC checkpoint and the PPO checkpoint initialized from that exact BC parent on each of the first three official levels: *Stereo Madness*, *Back on Track*, and *Polargeist*. The two auxiliary live evaluations use the same 25-attempt paired protocol at the 30-FPS evaluation cadence. The first uses *Stereo Madness Copy* (KRUTOYARBUS), a community copy of a later Level-1 segment, as a late-stage/ship-reachable proxy because official Level 1–3 attempts die before ship. The second uses one custom community level, *Stereo INSANE Nerfed* (XXcpta1n), as a single held-out layout within the supported mechanic distribution (no gravity inversion or other out-of-scope mechanics). This level was first accessed for this evaluation after the corpus and checkpoints were frozen and is therefore absent from the training data. In addition, a separate cadence-transfer probe reruns the historical frozen V7 PPO controller for 25 scored attempts on Stereo Madness Copy at 60 FPS on the optimized deploy-equivalent inference path (direct crop capture, exact CUDA preprocessing, FSQ `torch.compile`, transformer CUDA graph, GPU-resident context, no per-stage CUDA synchronization). Every paired evaluation first waits until the evaluator observes an in-level, alive state, then excludes one synchronization episode and retains the next 25 valid attempts. The exclusion prevents manual game resume or startup waiting time from inflating apparent survival. Game memory is used only to delimit alive/dead episodes, never as policy input. The primary endpoint is frames survived; when comparing 30-FPS and 60-FPS deployments, we use the recorded episode wall time from evaluation start to detected death because frame counts are not comparable across cadences. Reported variation within each frozen policy is the sample standard deviation across attempts. PPO-minus-parent-BC intervals are 95% percentile-bootstrap intervals using 50,000 resamples, with attempts resampled independently within each frozen policy. These intervals quantify deployment variability from capture and action timing, not controller-training uncertainty. Replication evidence is instead the three paired seed-level differences; attempts are never pooled across seeds and presented as independent training runs. All three

Table 2 Paired official-level live frame survival for three controller seeds. BC and PPO entries are mean $\pm$ sample standard deviation over 25 valid attempts. Each Δ row is PPO minus its exact parent BC, reported as mean [95% attempt-level bootstrap interval]. The three seed differences, rather than pooled attempts, are the replication evidence.

Seed	Policy	Stereo Madness	Back on Track	Polargeist
43	BC	120.2 $\pm$ 91.8	123.5 $\pm$ 81.5	52.4 $\pm$ 16.7
	PPO	280.1 $\pm$ 23.3	260.0 $\pm$ 55.5	65.2 $\pm$ 33.4
	Δ [95% CI]	159.9 [122.7, 195.4]	136.5 [98.8, 174.4]	12.8 [−0.2, 28.6]
44	BC	130.6 $\pm$ 76.0	111.1 $\pm$ 60.3	49.0 $\pm$ 16.5
	PPO	280.4 $\pm$ 33.5	195.9 $\pm$ 126.5	68.3 $\pm$ 44.1
	Δ [95% CI]	149.8 [116.8, 180.6]	84.8 [31.9, 139.6]	19.4 [3.0, 39.0]
45	BC	156.0 $\pm$ 100.2	120.2 $\pm$ 69.6	46.2 $\pm$ 9.2
	PPO	308.8 $\pm$ 68.6	209.4 $\pm$ 130.0	63.1 $\pm$ 32.9
	Δ [95% CI]	152.8 [106.7, 199.8]	89.2 [34.6, 147.2]	16.8 [5.7, 31.5]

controller pairs share one frozen tokenizer and world-model seed, so this is controller-seed replication rather than end-to-end training uncertainty. The earlier comparison against a separately reconstructed BC checkpoint is superseded and excluded from the reported control results. Deployment measurements use Desktop Duplication via `dxcam` directly over region $(x, y, s) = (660, 48, 1032)$, exact Sobel preprocessing to 64×64 , and `keyboard` space dispatch on PyTorch 2.11.0+cu126.

Latency and continuation protocols. Latency is measured on 5,000 consecutive frames of the final 60-FPS optimized deployment path, from capture through action update, with CUDA synchronization only at the end-to-end boundary. For the qualitative intervention, the world model is initialized from a fixed encoded four-frame prefix immediately before a spike. Greedy continuations differ only in the first conditioned action, jump or idle; all subsequent actions are idle.

External transition-latency protocol. To characterize computational design rather than control quality, we benchmark the released Pong checkpoints of IRIS and DIAMOND alongside the selected DashVMC checkpoint on the same RTX 2060 SUPER. Each model runs in a fresh process at batch size one with FP32 eager execution; after 30 warm-up transitions, 100 transitions are timed individually with CUDA synchronization at each boundary, and the full procedure is repeated twice. The measured operation follows each native imagination interface: DashVMC predicts a 64-token latent grid in parallel together with its death score; IRIS autoregressively predicts 16 tokens, decodes the observation, and predicts reward and termination; and DIAMOND performs three-step pixel diffusion and predicts reward and termination. The differing output representations make this an operational latency comparison, not a matched generation-quality or control benchmark.

5 Results

5.1 Frozen-system and live-control results

All component metrics below are reported at their checkpoint-selection points rather than as independent maxima. The selected tokenizer reaches 3.894×10^{-4} per-pixel validation MSE (34.10 dB PSNR), obtained by dividing its per-frame reconstruction SSE of 1.595 by 64^2 . On the unweighted validation distribution with its natural death frequency, the transformer reaches death precision/recall/F1 of 0.7337/0.8654/0.7941. Under the training-time full-vocabulary argmax metric, it reaches 29.74% token accuracy; low exact-token accuracy can coexist with useful local predictions in a 1,000-class grid, but it also bounds dream reliability. The selected BC validation accuracies for seeds 43–45 are 78.97%, 79.15%, and 79.15%; their selected PPO checkpoints survive 30.13, 29.95, and 29.96 of 45 steps on the shared fixed latent development contexts.

Table 2 shows a positive PPO-minus-parent-BC mean for all nine official-level seed pairs. Across the three controller seeds, the paired improvements are 154.2 ± 5.2 frames on *Stereo Madness*, 103.5 ± 28.7 on *Back on Track*, and 16.3 ± 3.3 on *Polargeist*, where the variation is the sample standard deviation across the three seed differences. Every attempt-level interval on the first two levels excludes zero. The smaller Polargeist improvement is positive for all three seeds, although seed 43’s deployment-variability interval marginally

Table 3 Paired auxiliary live frame survival under the same protocol as Table 2. Stereo Madness Copy is a late Level-1 community copy used as a ship-reachable segment proxy; Stereo INSANE Nerfed is one custom held-out layout within the supported mechanic distribution.

Seed	Policy	Stereo Madness Copy	Stereo INSANE Nerfed
43	BC	366.8 ± 172.8	104.8 ± 72.1
	PPO	435.6 ± 196.1	290.4 ± 48.6
	Δ [95% CI]	$68.8 [-33.3, 166.1]$	$185.5 [151.1, 217.6]$
44	BC	285.7 ± 153.5	102.1 ± 64.0
	PPO	495.8 ± 137.5	342.8 ± 80.8
	Δ [95% CI]	$210.1 [127.9, 285.5]$	$240.6 [200.8, 280.2]$
45	BC	299.4 ± 175.2	97.8 ± 57.9
	PPO	546.8 ± 94.1	266.7 ± 128.9
	Δ [95% CI]	$247.4 [167.6, 320.7]$	$168.8 [115.4, 223.9]$

includes zero; seeds 44 and 45 exclude it. Polargeist introduces yellow orbs requiring a second timed input while airborne, so these per-level results are not a controlled test of generalization. All three official live-evaluation levels occur in the offline corpus, but the learning agent has no live-game interaction before evaluation; the paired ordering is consistent with imagined policy optimization transferring to deployment, not evidence of held-out-level generalization. Table 3 reports the same exact-parent comparison on the two auxiliary layouts. All six auxiliary seed differences are positive. On Stereo Madness Copy, seed 43’s attempt-level interval includes zero while seeds 44 and 45 exclude it; the mean paired improvement across seeds is 175.4 ± 94.2 frames. On Stereo INSANE Nerfed, all three intervals exclude zero and the mean paired improvement is 198.3 ± 37.6 frames. We report the custom-level result as held-out transfer within the supported mechanic distribution, not as novel-mechanic or broad out-of-distribution generalization; the Level-1 copy is not counted as held-out content. The 60-FPS cadence probe on Stereo Madness Copy yields similar observed real-time survival: mean recorded episode wall time is 17.72 s at 60 FPS versus 17.79 s at 30 FPS, with a 95% bootstrap interval of $[-1.04, 0.89]$ s for the mean difference (60 FPS minus 30 FPS). Both conditions die in the same recurring 15–21 s band; frame counts roughly double under the higher cadence, as expected, so the cross-cadence comparison uses recorded time rather than frames. This supports practical 60-FPS deployment from 30-FPS training data on this layout, without claiming broad frame-rate adaptation as a separate contribution. Separate intervention runs suggest that skipping a few recurring blocking obstacles lets the policy reach later sections, but those assisted runs are not part of the controlled statistics.

5.2 Capture-to-action latency

On 5,000 consecutive frames of the final 60-FPS optimized path, end-to-end capture-to-action latency averages 12.018 ms (95% bootstrap interval for the mean $[11.982, 12.056]$ ms), with median 12.083 ms, p95 14.069 ms, and maximum 24.059 ms. Only 11/5,000 measurements exceed the 16.667-ms frame period. The reciprocal mean is 83.2 FPS of compute-equivalent throughput, while the p95 latency corresponds to 71.1 FPS; these are throughput summaries rather than claims of evaluated control at those cadences. A matched 300-frame before/after profile isolates the preprocessing change: replacing the hybrid CUDA-Sobel/CPU-resize path with the exact CUDA path reduces mean full-loop latency from 14.395 to 11.907 ms (17.3%) and p95 from 16.424 to 14.043 ms (14.5%).

Table 4 End-to-end latency for 5,000 consecutive frames of the final optimized path on an RTX 2060 at the 60-FPS deployment cadence. Deliberate cadence sleep is excluded.

Statistic	Value
Mean [95% bootstrap CI]	12.018 [11.982, 12.056] ms
Median	12.083 ms
p95	14.069 ms
Maximum	24.059 ms
Frames over 16.667 ms	11/5,000
Reciprocal-mean throughput	83.2 FPS

These timings differ from diagnostic evaluator logs that move context through CPU/NumPy and synchronize after each GPU stage. Official-level and 30-FPS auxiliary survival use that diagnostic path; the 60-FPS cadence probe and deployment latency use the optimized path. The two preprocessing implementations produced the same observation bytes in the identity gate and use the same frozen model/controller weights; the distinction is execution optimization rather than policy definition.

5.3 Batch-1 imagined-transition latency

Table 5 reports the two independent run means. All three released models execute within the RTX 2060 SUPER’s 8-GB memory budget at batch size one, so these measurements do not support a claim that IRIS or DIAMOND is undeployable on the target GPU. DashVMC’s native imagined transition averages 11.05 ms, compared with 49.30 ms for DIAMOND and 177.83 ms for IRIS, corresponding to $4.5\times$ and $16.1\times$ lower latency, respectively. Only the DashVMC transition itself falls within a 33.3-ms frame period on this hardware, although the external agents can still infer real-environment actions without executing these generative transitions.

Table 5 Batch-1 native imagination-transition latency on an RTX 2060 SUPER. Values are synchronized run means in milliseconds. The models produce different representations and were not trained on a common dataset; this table compares operational cost, not quality or control.

Model	Run 1	Run 2	Mean
DashVMC	11.04	11.06	11.05
DIAMOND	50.13	48.46	49.30
IRIS	176.94	178.73	177.83

5.4 Action-conditioned continuations

World-model generation is not capped by the 45-step PPO horizon: each predicted grid and human-selected action can be fed back into the rolling context, so the interactive loop can continue for an arbitrary number of steps unless predicted death or an optional user limit stops it. Recorded examples can remain visually and mechanically plausible through substantial stretches, including height transitions and avatar transformations, before drift produces repeated motifs, empty segments, or impossible geometry. We therefore treat this as qualitative, open-ended level continuation rather than indefinitely coherent generation. Additional successes and failure cases are available on the [project page](#).

MIRA emphasizes that action conditioning does not itself establish action adherence and evaluates commanded-action recoverability from generated video (Hu et al., 2026). Our matched-prefix test is a smaller qualitative intervention check rather than a population-level adherence metric. Figure 2 tests action sensitivity while holding the visual context fixed. The jump-conditioned branch clears the spike and lands; the idle-conditioned branch reaches a death score of 0.74 at $t + 5$ and terminates. Decoding is used only for inspection, not controller training.

5.5 Design diagnostics

Table 6 condenses engineering variants explored before freezing the system. These diagnostics span successive development stacks; except for the quantified BC timing comparison, they are local selection records rather than balanced, directly comparable statistical ablations. They explain design choices but do not establish general architectural rankings.

Historical run records and code snapshots support these pre-freeze diagnostics; the exact final-system claims above use the frozen artifacts specified in the evaluation protocol.

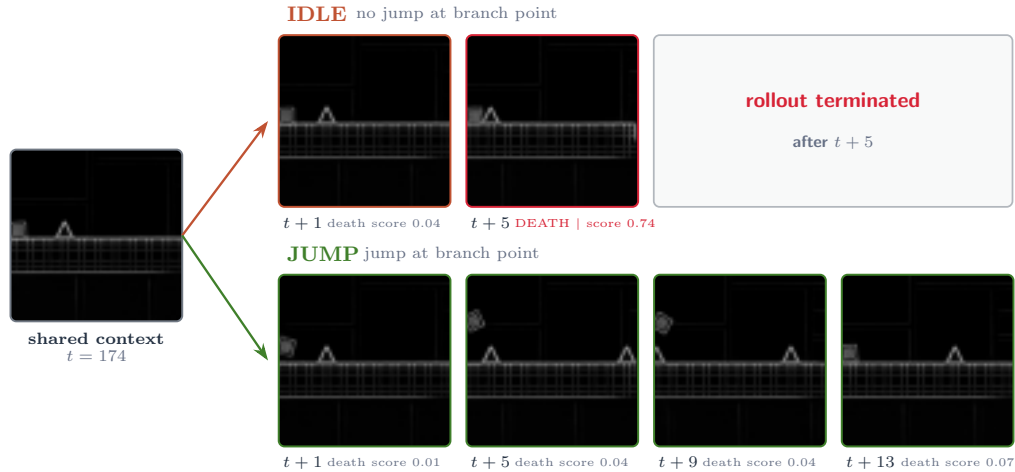


Figure 2 Greedy continuations from one reconstructed four-frame prefix; only the first future action differs. At $t + 1, t + 5, t + 9, t + 13$, the jump branch clears the spike. The idle branch terminates at $t + 5$ (death score 0.74), so later frames are not generated.

Table 6 Pre-freeze design diagnostics from successive development stacks. “Live” denotes real-game progress in local evaluation.

Variant	Observed outcome	Decision
PPO without BC (earlier stack)	Same latent plateau, about 2,000 more iterations (≈ 6 A100 hours); BC takes about 5 minutes.	Keep BC.
AdaLN action conditioning (Peebles and Xie, 2023)	Canonical zero-gate initialization stalled; bias-1 gates plus QK-normalization (Esser et al., 2024) trained stably and reached the best development CPC. The trial co-varied other components and did not establish a live advantage over action tokens.	Use action tokens.
Transformer width 512	Better CE/contrastive metrics, no live gain, slower training.	Use width 384.
MLP on $\mathbf{h}_t$ only	PPO was 2–5 $\times$ faster with a smaller train–eval gap, but local live progress was worse.	Keep grid input.

6 Limitations

DashVMC is a single-game study with binary actions and deterministic layouts. The tokenizer and world model remain one frozen training seed; seeds 43–45 replicate only BC and PPO controller training conditional on that shared representation and dynamics model. Three controller seeds establish repeatability of the observed direction within this frozen system but are too few to characterize broad training uncertainty, and survival time is not level completion. Attempt-level bootstrap intervals quantify deployment variability for each frozen pair, not uncertainty over controller seeds or end-to-end systems. The auxiliary custom-level evaluation uses a single held-out community layout; it supports within-mechanic transfer rather than broad generalization. The Level-1 copy evaluates later-stage survival but does not isolate ship mode with explicit mode labels. The 60-FPS cadence comparison is a single-level, single-policy probe with $n = 25$ and also changes from the diagnostic to the optimized execution path; it is an operational deployment probe, not an isolated or broad study of frame-rate adaptation. Each reported PPO controller is compared with the exact retained BC checkpoint that initialized it, but the live suite has only 25 valid attempts per frozen policy-level. Because no other world-model agent is available under the same captured-pixel, non-paused Geometry-Dash protocol, the results are a controlled within-system comparison rather than a state-of-the-art benchmark. IRIS and DIAMOND were not retrained on the DashVMC corpus, and their released recurrent policies can infer real-environment actions without generating the transitions timed in Table 5. That comparison therefore motivates the computational design of DashVMC’s imagination path but establishes neither that the alternatives could not control Geometry Dash at the required cadence nor that DashVMC would outperform

them after matched training and tuning.

The black-box agent interface should be distinguished from the experimental infrastructure. Restricting the agent to captured pixels and keyboard actions is central to the deployment setting, but requiring every interaction to occur through a visible, non-paused desktop application is not intrinsically necessary for studying pixel-based control. This coupling made data collection and repeated evaluation proceed at wall-clock speed, limited straightforward acceleration and parallelization, and complicated automated resets, debugging, and reproducible measurement. For research focused primarily on representation learning, world modeling, or control, we therefore recommend using a headless or step-controllable environment while deliberately withholding privileged state from the agent. Live interaction with the original application is most valuable as a final deployment test, or when satisfying the real-time capture-to-action contract is itself the research question.

The Geometry-Dash evidence for Equation 1 is a single-run, same-architecture comparison against uniform smoothing, not a replicated hard-CE ablation; it supports the local choice but neither quantifies uncertainty nor isolates transfer to the final system. In a matched IRIS/Pong diagnostic with five seeds per condition, annealed SLS underperformed CE in final return (12.14 ± 10.82 versus 15.68 ± 4.96) and doubled the tail-failure rate (40% versus 20%), despite lower drawdown and return volatility and a transient advantage during epochs 250–420. We therefore treat any benefit as specific to the structured FSQ geometry and Geometry-Dash setting rather than claim a general discrete-world-model improvement. The open-ended decoded continuations are qualitative: coherence is not scored, and the curated examples do not establish how frequently a rollout remains plausible. An exact multi-step match to a recorded continuation would also conflate predictable local dynamics with geometry that enters from outside the observed seed and is therefore not identifiable from the prefix. We leave visibility-conditioned multi-step fidelity evaluation to future work; autoregressive drift can yield repetitions, empty stretches, or impossible states, and the generated continuations are neither editor-valid nor guaranteed playable.

Successive development-time dataset doublings produced visibly more coherent, higher-quality rollouts without an observed plateau, suggesting that the present model remains data-limited; these runs were not controlled end-to-end scaling experiments. Although the system is learned from a small offline corpus, we do not report a data-scaling curve or matched larger-data baseline. The results therefore demonstrate viability under a constrained data budget, not general sample efficiency. Under matched reconstruction constraints, controlled iFSQ image-generation experiments find that autoregressive models converge faster early in training, whereas diffusion models reach a higher generation-quality ceiling with sufficient compute (Lin et al., 2026). These findings motivate—but do not validate—our preference for faster early convergence under a small data budget; they do not establish that the trade-off transfers to Sobel-preprocessed temporal prediction or downstream control, nor that autoregression is universally preferable.

The reported 12.0-ms profile covers this hardware and 64-pixel edge observations; RGB input, stochastic environments, larger action spaces, or weaker devices require new accuracy–latency measurements.

7 Conclusion

DashVMC shows that a lightweight and specialized world model can support both low-latency imagined transitions and live reactive control in a timing-sensitive game. Across three controller seeds, PPO improves mean live survival over its exact retained parent BC on all three official levels for every seed and preserves that direction on a late Level-1 copy and one custom held-out community level, while the optimized capture-to-action path sustains a 60-FPS live cadence on a consumer GPU despite 30-FPS training data. This repeated paired ordering is consistent with imagined PPO optimization transferring to live deployment conditional on one frozen tokenizer and world model; it is not full end-to-end training replication. A matched-prefix intervention shows that the model’s visual future responds to the tested action change, and interactive examples show that those dynamics can produce open-ended, sometimes plausible level continuations, albeit without indefinite coherence guarantees.

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References

- Eloi Alonso, Adam Jelley, Vincent Micheli, Anssi Kanervisto, Amos Storkey, Tim Pearce, and François Fleuret. Diffusion for world modeling: Visual details matter in atari. In *Advances in Neural Information Processing Systems*, volume 37, 2024.
- Bowen Baker, Ilge Akkaya, Peter Zhokhov, Joost Huizinga, Jie Tang, Adrien Ecoffet, Brandon Houghton, Raul Sampedro, and Jeff Clune. Video pretraining (VPT): Learning to act by watching unlabeled online videos. *arXiv preprint arXiv:2206.11795*, 2022.
- Maxime Burchi and Radu Timofte. TWISTER: Learning transformer-based world models with contrastive predictive coding. In *International Conference on Learning Representations*, 2025.
- Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorber, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *International Conference on Machine Learning*, 2024.
- Junliang Guo, Yang Ye, Tianyu He, Haoyu Wu, Yushu Jiang, Tim Pearce, and Jiang Bian. MineWorld: A real-time and open-source interactive world model on Minecraft. *arXiv preprint arXiv:2504.08388*, 2025.
- David Ha and Jürgen Schmidhuber. Recurrent world models facilitate policy evolution. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 2025.
- Xianglong He, Chunli Peng, Zexiang Liu, Boyang Wang, Yifan Zhang, Qi Cui, Fei Kang, Biao Jiang, Mengyin An, Yangyang Ren, Baixin Xu, Hao-Xiang Guo, Kaixiong Gong, Size Wu, Wei Li, Xuchen Song, Yang Liu, Yangguang Li, and Yahui Zhou. Matrix-Game 2.0: An open-source real-time and streaming interactive world model. *arXiv preprint arXiv:2508.13009*, 2025.
- Todd Hester, Matej Vecerik, Olivier Pietquin, Marc Lanctot, Tom Schaul, Bilal Piot, Dan Horgan, John Quan, Andrew Sendonaris, Gabriel Dulac-Arnold, Ian Osband, John Agapiou, Joel Z. Leibo, and Audrunas Gruslys. Deep q-learning from demonstrations. In *AAAI Conference on Artificial Intelligence*, 2018.
- Anthony Hu, Václav Volhejn, Adrien Ramanana Rahary, Chris Mulder, Aditya Makkar, Alyx Liao, Amélie Royer, Manu Orsini, Adam Jelley, Eloi Alonso, Florian Laurent, Fredrik Norén, James Swingos, Jan Hünemann, Kent Rollins, Lucas Hosseini, Matthieu Le Cauchois, Maxim Peter, Pim de Witte, Tim Brown, Vincent Micheli, Moritz Böhle, Gabriel de Marmiesse, Viktoriia Sharmanska, Lucia Specia, Michael Black, and Patrick Pérez. Multiplayer interactive world models with representation autoencoders. *arXiv preprint arXiv:2607.05352*, 2026.
- Yann LeCun. A path towards autonomous machine intelligence. *OpenReview*, 2022. Version 0.9.2.
- Ted Li and Sean Rafferty. Playing Geometry Dash with convolutional neural networks. Stanford CS231N course report, 2017.
- Bin Lin, Zongjian Li, Yuwei Niu, Kaixiong Gong, Yunyang Ge, Yunlong Lin, Mingzhe Zheng, JianWei Zhang, Miles Yang, Zhao Zhong, Liefeng Bo, and Li Yuan. iFSQ: Improving FSQ for image generation with 1 line of code. *arXiv preprint arXiv:2601.17124*, 2026.
- Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *International Conference on Computer Vision*, pages 2980–2988, 2017.
- Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. LeWorldModel: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv preprint arXiv:2603.19312*, 2026.
- Fabian Mentzer, David Minnen, Eiríkur Agustsson, and Michael Tschannen. Finite scalar quantization: VQ-VAE made simple. In *International Conference on Learning Representations*, 2024.

- Vincent Micheli, Eloi Alonso, and François Fleuret. Transformers are sample-efficient world models. In *International Conference on Learning Representations*, 2023.
- Vincent Micheli, Eloi Alonso, and François Fleuret. Efficient world models with context-aware tokenization. In *International Conference on Machine Learning*, 2024.
- William Peebles and Saining Xie. Scalable diffusion models with transformers. In *International Conference on Computer Vision*, 2023.
- John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. *arXiv preprint arXiv:1506.02438*, 2015.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.
- Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the Inception architecture. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 2818–2826, 2016.
- Hugues Van Assel, Mark Ibrahim, Tommaso Biancalani, Aviv Regev, and Randall Balestriero. Joint embedding vs reconstruction: Provable benefits of latent space prediction for self supervised learning. *arXiv preprint arXiv:2505.12477*, 2025.
- Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.
- Zaishuo Xia, Yukuan Lu, Xinyi Li, Yifan Xu, and Yubei Chen. Clone deterministic 3d worlds with geometrically-regularized world models. *arXiv preprint arXiv:2510.26782*, 2025.
- Tongda Xu, Wendi Zheng, Jiajun He, José Miguel Hernández-Lobato, Yan Wang, Ya-Qin Zhang, and Jie Tang. Training-free vector quantization via gaussian VAEs. *arXiv preprint arXiv:2512.06609*, 2025.